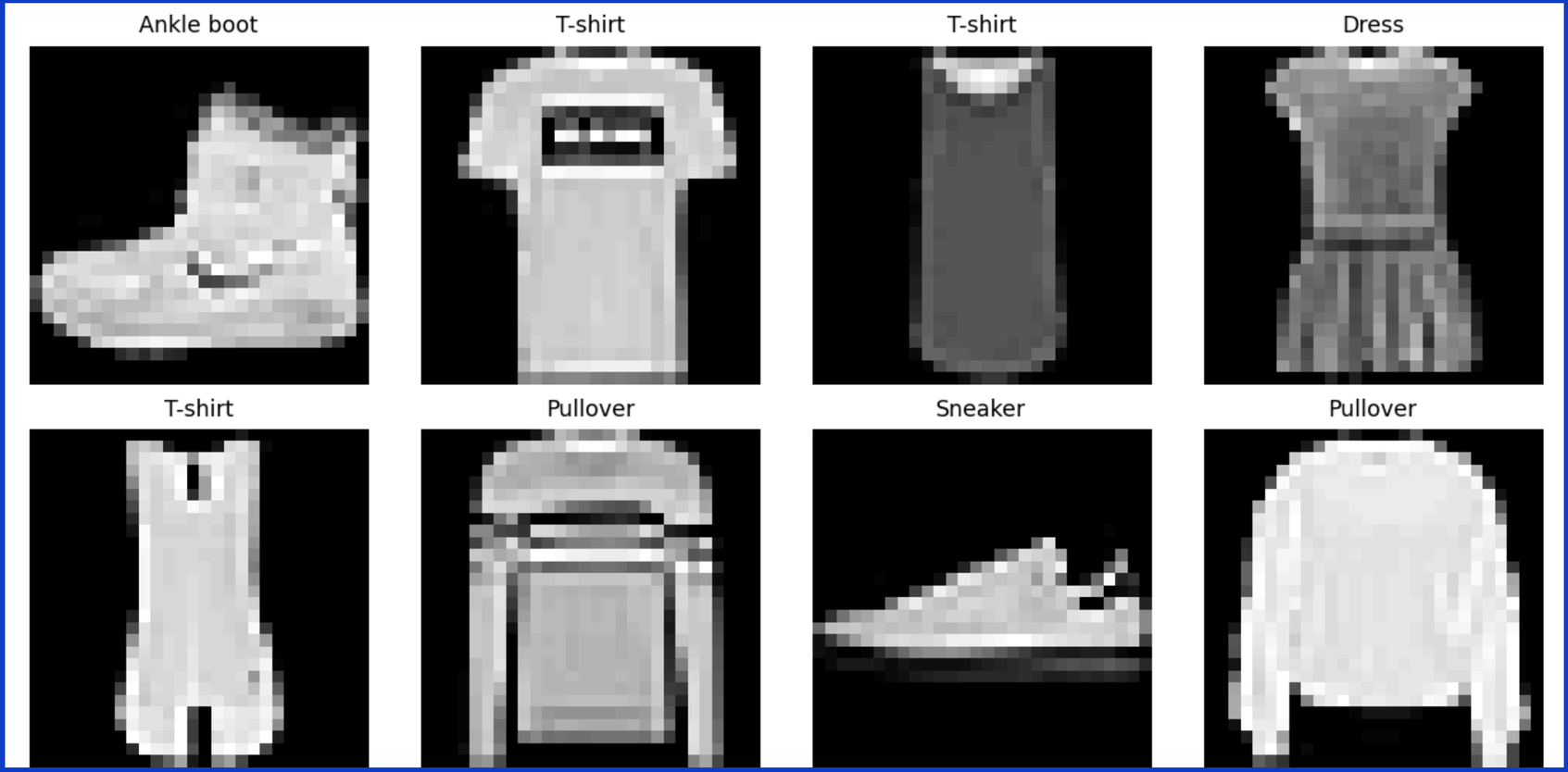
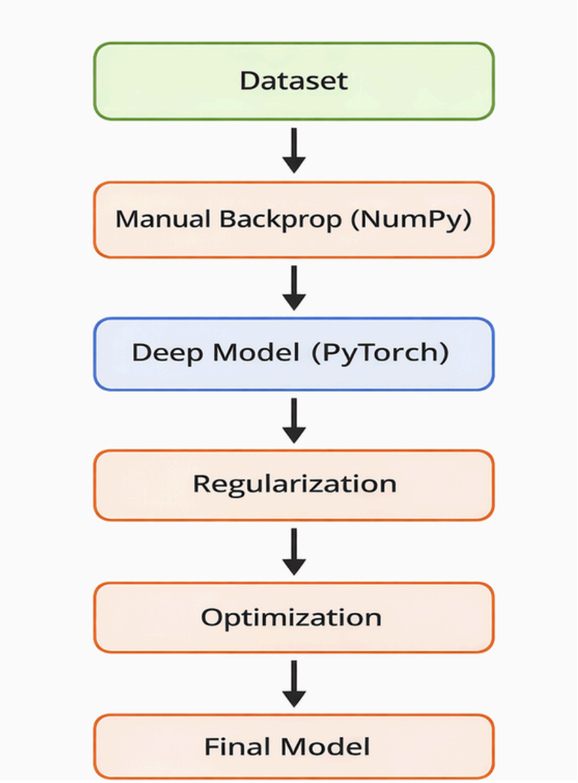
Building and Improving Neural Networks on FashionMNIST

FashionMNIST Classification



Model Development Pipeline



Dataset → Input data

Manual Backprop → Understand gradients

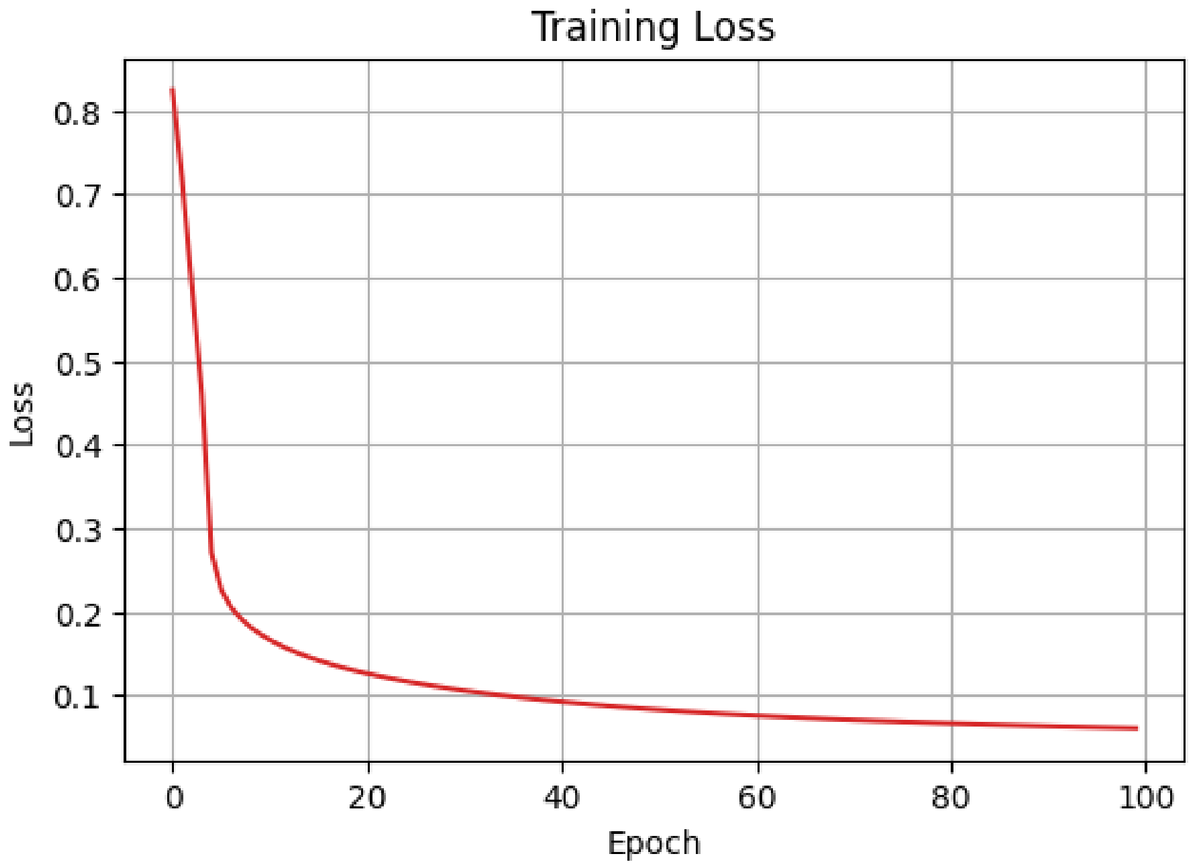
Deep Model → Learn complex patterns

Regularization → Reduce overfitting

Optimization → Improve convergence

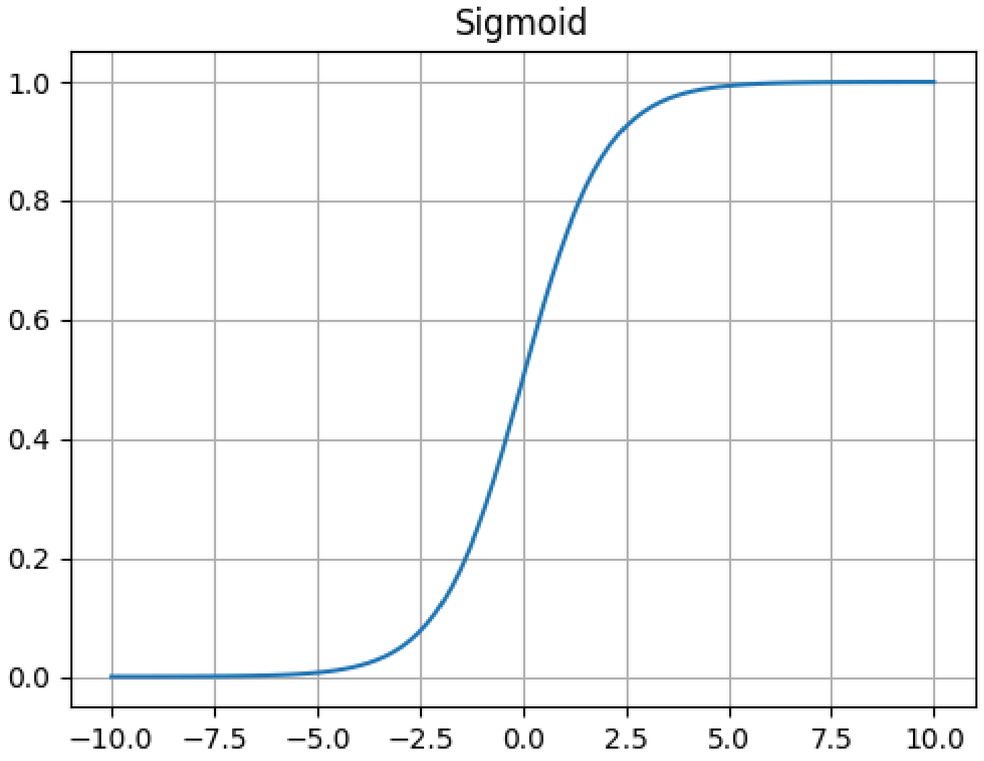
Final Model → Best performance

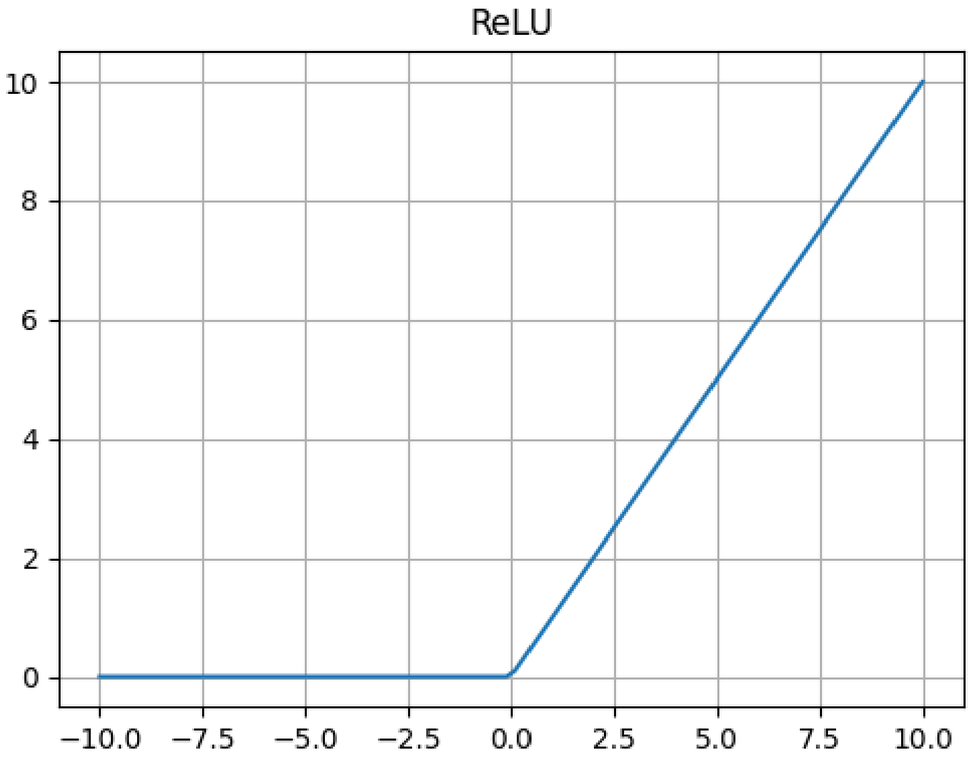
Training Process (Loss Reduction)



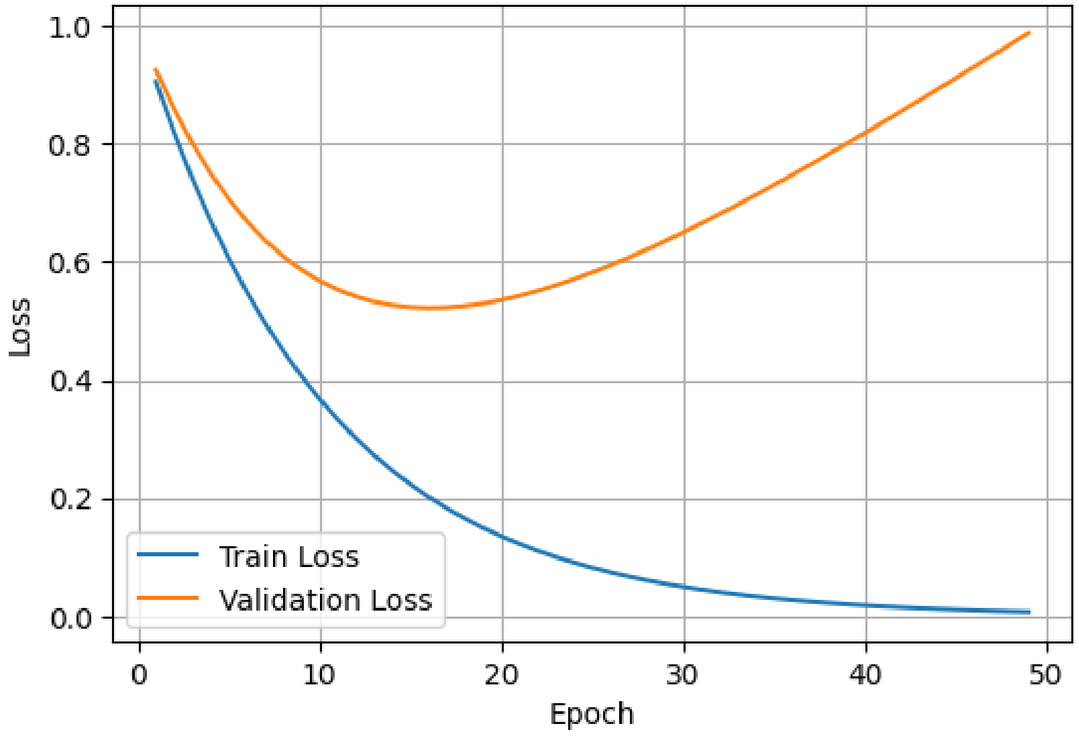
Loss decreases over epochs

Activation Functions: Sigmoid vs ReLU





Overfitting and Regularization

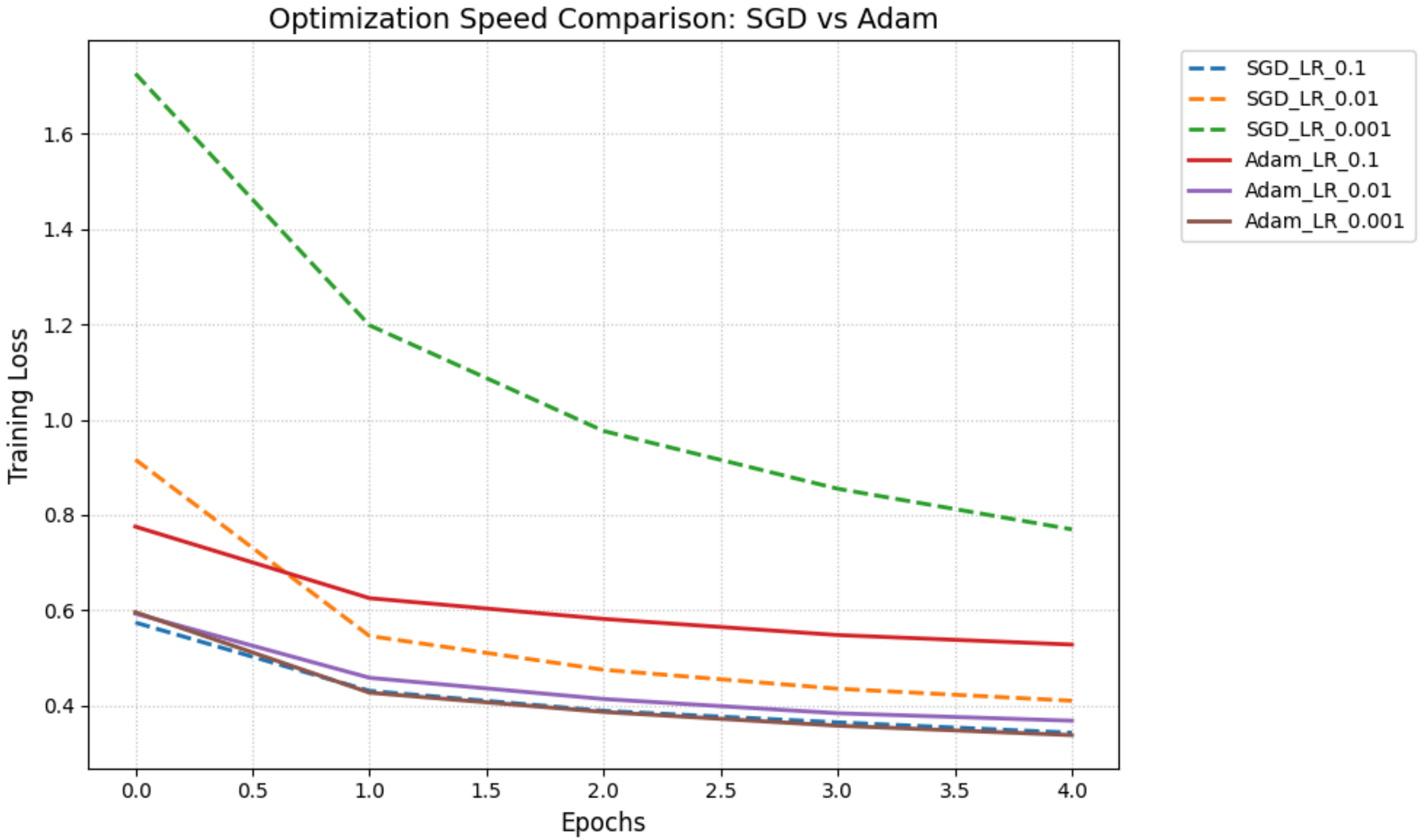


L2 Regularization

Dropout

Early Stopping

Optimization and Training Stability



Results and Key Insights

NumPy Model → 97.9%
PyTorch Model → 97.6%

Gradient flow matters

Activation choice matters

Regularization improves generalization

Optimization improves convergence